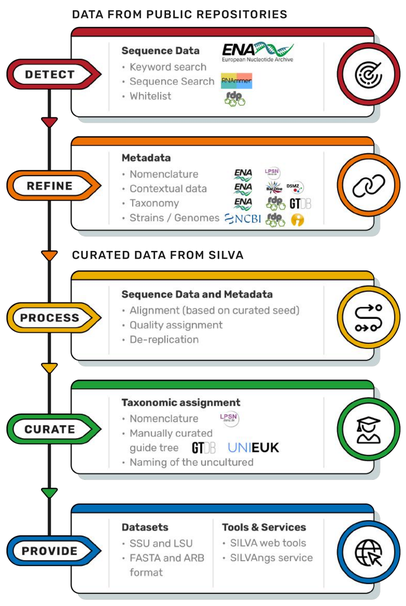
Joint forces of three ELIXIR Core Data Resources:

Building the new biodata infrastructure DSMZ Digital Diversity

Martinique Frentrup, Lorenz C. Reimer, Antje Jäde, Isabel Schober, Jan Gerken and the Digital Diversity Team

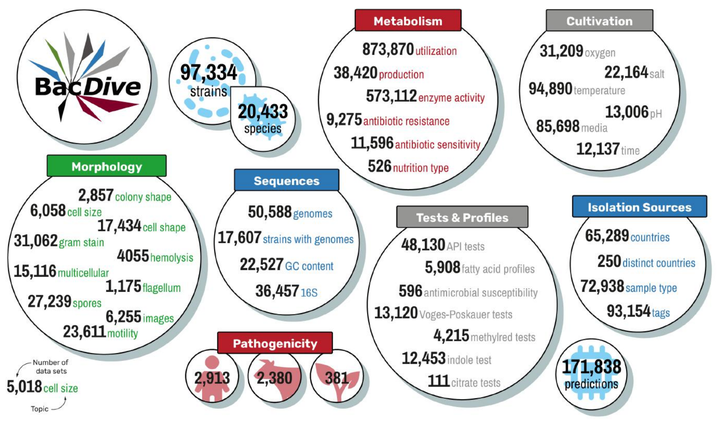
silva

SILVA is a comprehensive resource for quality checked, regularly updated and aligned ribosomal RNA sequence data for all three domains of life and offers several to tools for data analysis.



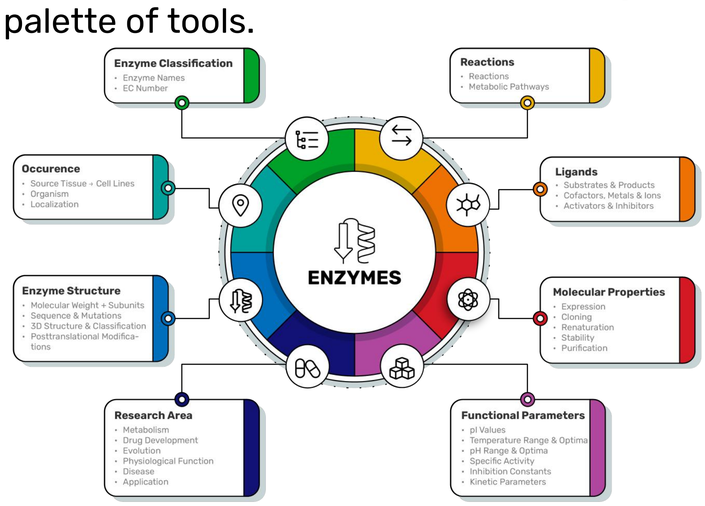
BacDive

The Bacterial Diversity Database Bac*Dive* is the worldwide largest database for standardized bacterial phenotypic information.



BRENDA

The BRENDA Enzyme Database is the world's most comprehensive enzyme information repository and offers a large palette of tools.



TYGS

The Type (Strain) Genome Server is a user-friendly high-throughput web server for genome-based prokaryote taxonomy.

LPSN .dsmz.de

Comprehensive information on The List of Prokaryotic names with Standing in Nomenclature with weekly updates.

MediaDive

Media*Dive* is a manually curated collection of cultivation media and growth conditions for all domains of life.

StrainInfo

StrainInfo is a service providing a resolution of microbial strain IDs by storing culture collection numbers and culture-associated data.

DSMZ CellDive

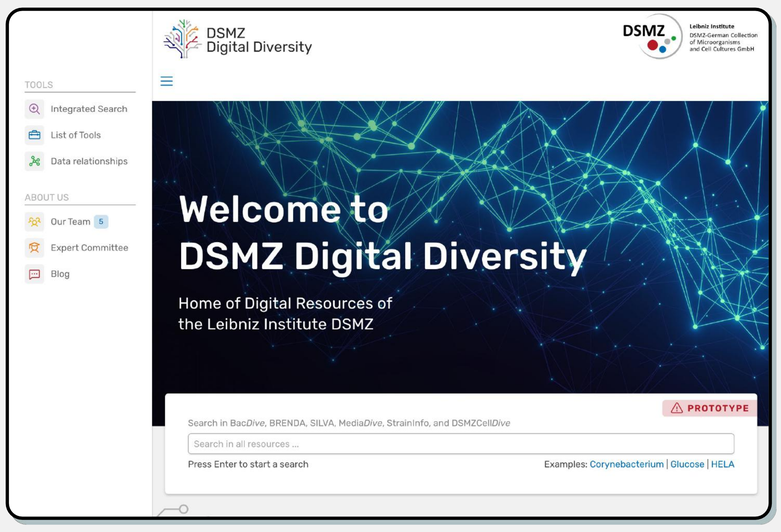
DSMZCellDive helps with selecting suitable model cell lines by providing insights into gene expressions and more.

PhageDive

Phage*Dive* is a specific database for bacteriophages and achaeal viruses, gathering experimental data and metadata.

THE DIGITAL DIVERSITY HUB

This is where everything comes together

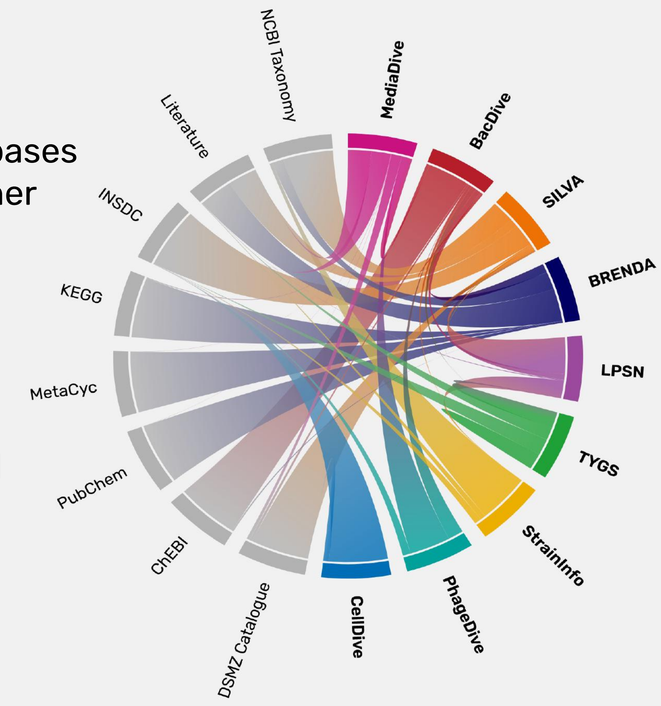


Our data is already connected...

The chart on the right shows how our databases are connected with each other and with other important international databases

... but not yet integrated

To integrate the data one has to understand them and draw the correct relationships. Our data integration team will develop a knowledge graph to tackle this.



What we plan to do

Digital Diversity Hub

- Federated search across all databases
- Integrated data profile

Plant Virus Database

- Providing tools and data on plant viruses

Knowledge Graph

- Build unified ontology
- Knowledge graph for all resources & UI

Metabolic Pathways

- Toolbox for automatically drawn pathways
- Organism-specific

Annotation Suite

- Improve annotation for BacDive, BRENDA & LPSN
- Text-mining assistance

Isolation Sources

- Standardize isolation sources
- Semi-automatic annotation (AI)

Join the team

From left to right: Lisa Jeske, Boyke Bunk, Isabel Schober, Markus Göker, Katrin Dümmer, Antje Jäde, Julia Koblitz, Heike Freese, Christian-Alexander Dudek, Artur Lissin, Joaquim Sarda, Jan Gerken, Lorenz Reimer, Julius Witte
Missing in the picture: Christian Ebeling, Martinique Frentrup, Robin Goldmann, Yeliz Sandikci, Anja Tietz

DSMZ Digital Diversity

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